

Report for Mandatory Assignment 5 (Replication)

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Github Link:

https://github.com/Asger124/Distributed/tree/Asger_tcp/Replication

- **Introduction**

- The system is a distributed auction system, that has a client-server architecture. The nodes in the system communicate through grpc. The system consists of 3 server nodes which are running on port 5001, 5002, and 5003. Server nodes contain all necessary information about the auction.
- It is possible to connect multiple clients in the system, and they will run on port $6000 + n$. Each client can make multiple bids, and query the servers for information about the auction
- The system is based on 3 replica servers and is resilient to crashes of 2/3 server nodes.

- **Architecture**

- The system is based on “Active replication”, and act as described below:

1. Request

- The clients send requests which are multicast to replica servers. They do not issue the next request until they get a response.

2. Coordination

- Since every client is connected to all replica servers the requests will be delivered in the same total order across the replica servers.

3. Execution

- Every replica (that is not crashed) in the system will execute requests deterministically.

4. Response

- Each replica(not crashed) will send its response to the requesting client.

- **Correctness 1**

- The implementation achieves sequential consistency due to the points above, and it satisfies the property that: the result of any execution is the same as if the operations of all the processors were executed in some sequential order, and the operations of each individual processor appear in this sequence in the order, of replica servers receiving requests. Meaning that all processes will observe the same order, and the results will be consistent across all processes.

- **Correctness 2**

- The protocol is correct in the absence and presence of failures. Since all client's multicast to all replica servers, the server nodes will always be up to date and the clients will receive correct information in response, even when one or more replica server nodes are down. If all replica server nodes fail, no real information about the auction will be present in the response – only a message stating that all servers are down.